

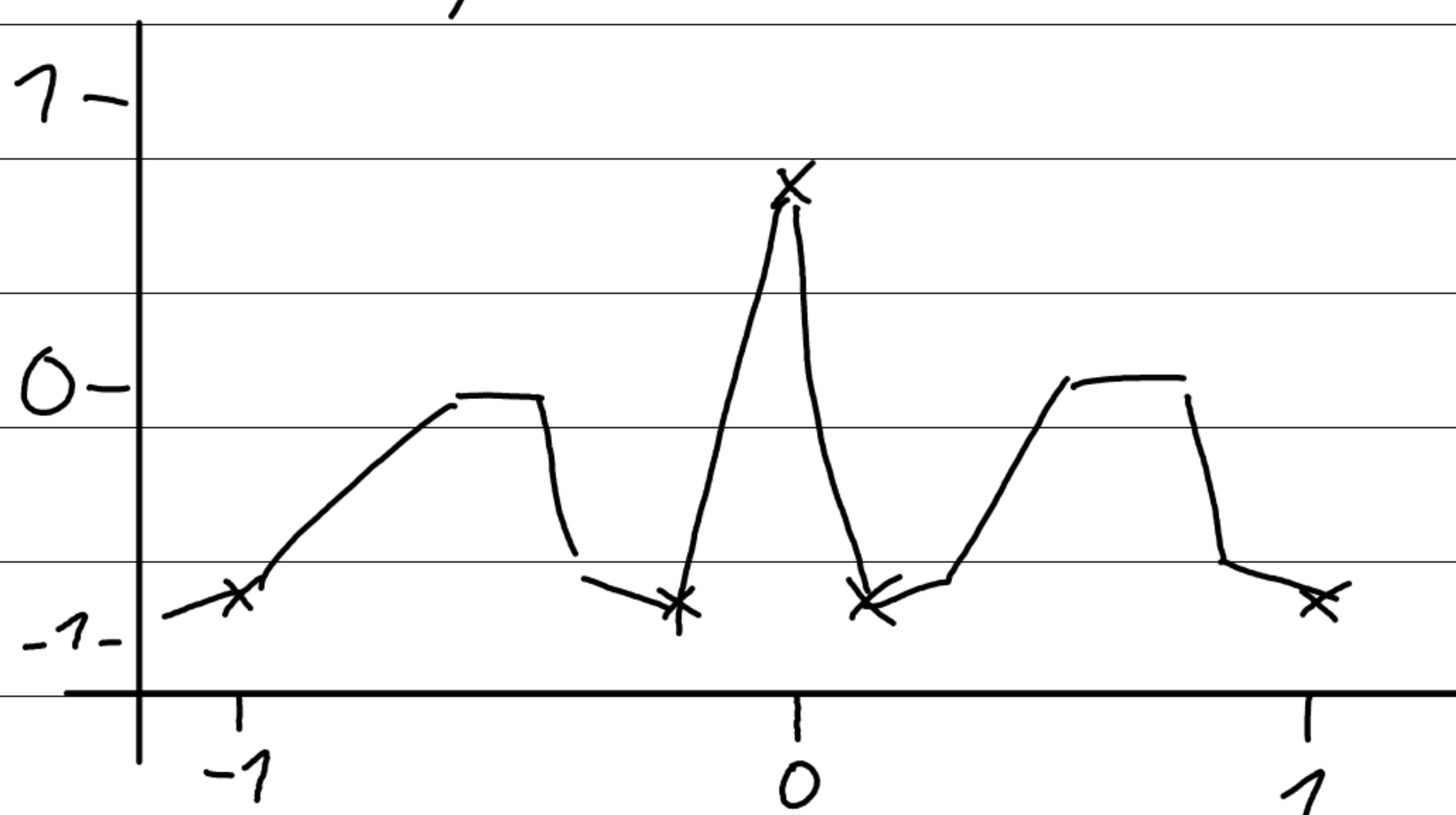
4 Deep neural networks

4.1 e.g. $x=0 \leadsto y=-1 \leadsto y'=0.6$

$x=-1, \leadsto y=0 \leadsto y'=-0.9$

$x=1,$

$x=0.2, x=-0.2$



4.2 4 hyperparameters: D_1, D_2, D_3, D_4 , U layers

$$\begin{aligned} 4.3 \text{ Is } & \text{ReLU}[\beta_1 + \lambda_1 \Omega_1 \text{ReLU}[\beta_0 + \lambda_0 \Omega_0 x]] \\ &= \lambda_0 \lambda_1 \text{ReLU}[\lambda_0 \lambda_1 \beta_1 + \Omega_1 \text{ReLU}[\lambda_0 \beta_0 + \Omega_0 x]]; \\ \text{using } & \text{ReLU}[az] = a \text{ReLU}[z] ; \lambda_0, \lambda_1 > 0 \end{aligned}$$

$$\begin{aligned} & \lambda_0 \lambda_1 \text{ReLU}[\lambda_0 \lambda_1 \beta_1 + \Omega_1 \text{ReLU}[\lambda_0 \beta_0 + \Omega_0 x]] \\ &= \text{ReLU}[\beta_1 + \lambda_0 \lambda_1 \Omega_1 \text{ReLU}[\lambda_0 \beta_0 + \Omega_0 x]] \\ &= \text{ReLU}[\beta_1 + \lambda_1 \Omega_1 \text{ReLU}[\beta_0 + \lambda_0 \Omega_0 x]] \quad \square \end{aligned}$$

$$4.4 \quad D_i = 5, D_0 = 4, U = 3, D_1 = 20, D_2 = 10, D_3 = 7$$

$$h_1 = a[\beta_0 + \Omega_0 x], \beta_0: 20 \times 1, \Omega_0: 20 \times 5$$

$$h_2 = a[\beta_1 + \Omega_1 h_1], \beta_1: 10 \times 1, \Omega_1: 10 \times 20$$

$$h_3 = a[\beta_2 + \Omega_2 h_2], \beta_2: 7 \times 1, \Omega_2: 7 \times 10$$

$$y = \beta_3 + \Omega_3 h_3, \beta_3: 4 \times 1, \Omega_3: 4 \times 7$$

$$y = \beta_3 + \Omega_3 a[\beta_2 + \Omega_2 a[\beta_1 + \Omega_1 a[\beta_0 + \Omega_0 x]]]$$

$$4.5 \quad \text{Width: } D = 30$$

$$\text{Depth: } U = 20 \quad (\text{hidden layers})$$

$$4.6 \quad D_i = 1, D_0 = 1$$

$$U = 10, D = 10$$

$$\begin{array}{c} 10 \\ 0 \rightarrow 0 \rightarrow \dots 0 \\ 0 \rightarrow \dots 0 \\ \vdots \\ 0 \rightarrow \dots 0 \\ 0 \rightarrow \end{array}$$

weights

100 more if we have one layer $n_{ul}: 10 \times 10$

~~100 more: 2 + 2 + 2 + 2~~

$$\begin{array}{c} 9 \text{ times} \\ \underbrace{11 \quad 11 \quad 11} \\ \rightarrow \quad \dots \quad \rightarrow \end{array}$$

if we increase the width by 1:

$$1 + 9 \cdot (11^2 - 10^2) + 1 = 191$$

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 input between new old output
 hidden layers

$$4.7 \quad y = \phi_0 + \phi_1 a [\theta_{10} + \theta_{11} x] + \phi_2 a [\theta_{20} + \theta_{21} x] + \phi_3 a [\theta_{30} + \theta_{31} x]$$

$$\phi_0 = 0, \quad \phi_1 = 1, \quad \theta_{10} = 0, \quad \theta_{11} = 1$$

$$\phi_2 = -1, \quad \theta_{20} = 0, \quad \theta_{21} = -1$$

$$\phi_3 = 0, \quad \theta_{30} = 0, \quad \theta_{31} = 0$$

$$\Rightarrow y = \text{ReLU}(x) - \text{ReLU}(-x) = x$$

$$\text{(alternatively: } y = a + \text{ReLU}(x-a) - \text{ReLU}(x-b)$$

$$\begin{aligned} \phi_0 &= a, \quad \theta_{11} = 1, \quad \theta_{10} = -a, \\ \phi_1 &= 1, \quad \phi_2 = -1, \\ \theta_{20} &= -b, \quad \theta_{21} = 1 \end{aligned}$$

4.8

$$h_1(x) = a [\theta_{10} + \theta_{11} x] = \max(0, x - \frac{1}{6})$$

$$h_2(x) = a [\theta_{20} + \theta_{21} x] = \max(0, x - \frac{2}{6})$$

$$h_3(x) = a [\theta_{30} + \theta_{31} x] = \max(0, -x + \frac{4}{6})$$

$$\Delta f = (\text{slope}) \times (\text{length})$$

in each interval, the active units are:

$$[0, \frac{1}{6}[: h_3 \Rightarrow h'_1 = h'_2 = 0, \quad h'_3 = -1 \Rightarrow f'(x) = -\phi_3 > 0$$

$$\Rightarrow \text{slope: } s_1 = 6$$

$$[\frac{1}{6}, \frac{2}{6}[: h_1, h_3 \Rightarrow h'_2 = 0, \quad h'_1 = 1, \quad h'_3 = -1 \Rightarrow f'(x) = \phi_1 - \phi_3 < 0$$

$$\Rightarrow \text{slope: } s_2 = -6$$

$$[\frac{2}{6}, \frac{4}{6}[: h_1, h_2, h_3 \Rightarrow h'_1 = h'_2 = 1, \quad h'_3 = -1 \Rightarrow f'(x) = \phi_1 + \phi_2 - \phi_3 > 0$$

$$\Rightarrow \text{slope: } s_3 = 3$$

$$[\frac{4}{6}, 1]: h_1, h_2 \Rightarrow h'_1 = h'_2 = 1, \quad h'_3 = 0 \Rightarrow f'(x) = \phi_1 + \phi_2 < 0$$

$$\Rightarrow \text{slope: } s_4 = -3$$

$$\text{Thus: } \phi_3 = -6 \Rightarrow \phi_1 + 6 = -6 \Rightarrow \phi_1 = -12$$

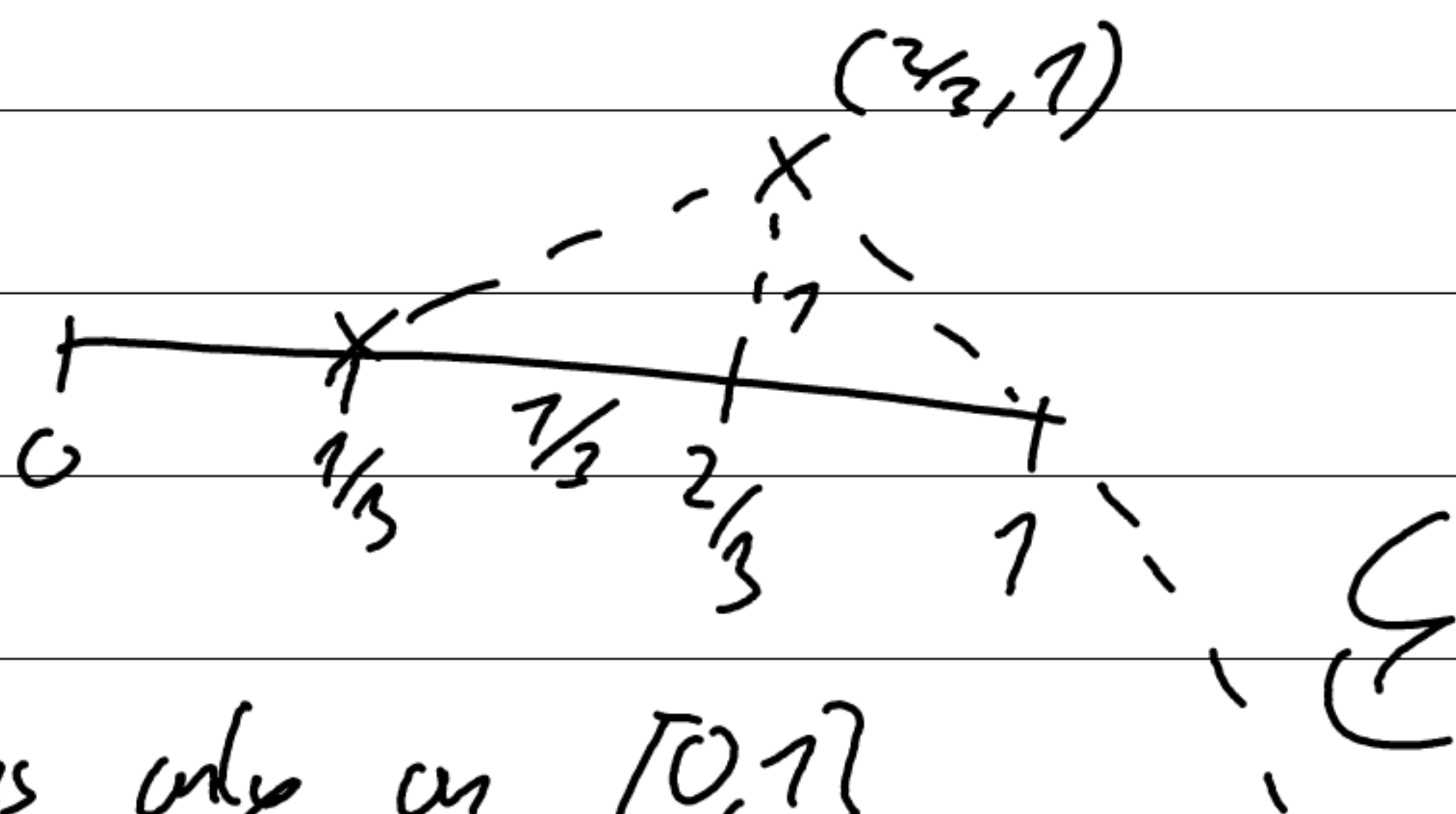
$$\Rightarrow -12 + \phi_2 = -3 \Leftrightarrow \phi_2 = 9$$

$$f(0) = \phi_0 + \phi_3 h_3 = \phi_0 - 6 \cdot \frac{4}{6} \Leftrightarrow \phi_0 = 4$$

Composing with itself gives $4 \cdot 4 = 16$ linear regions.

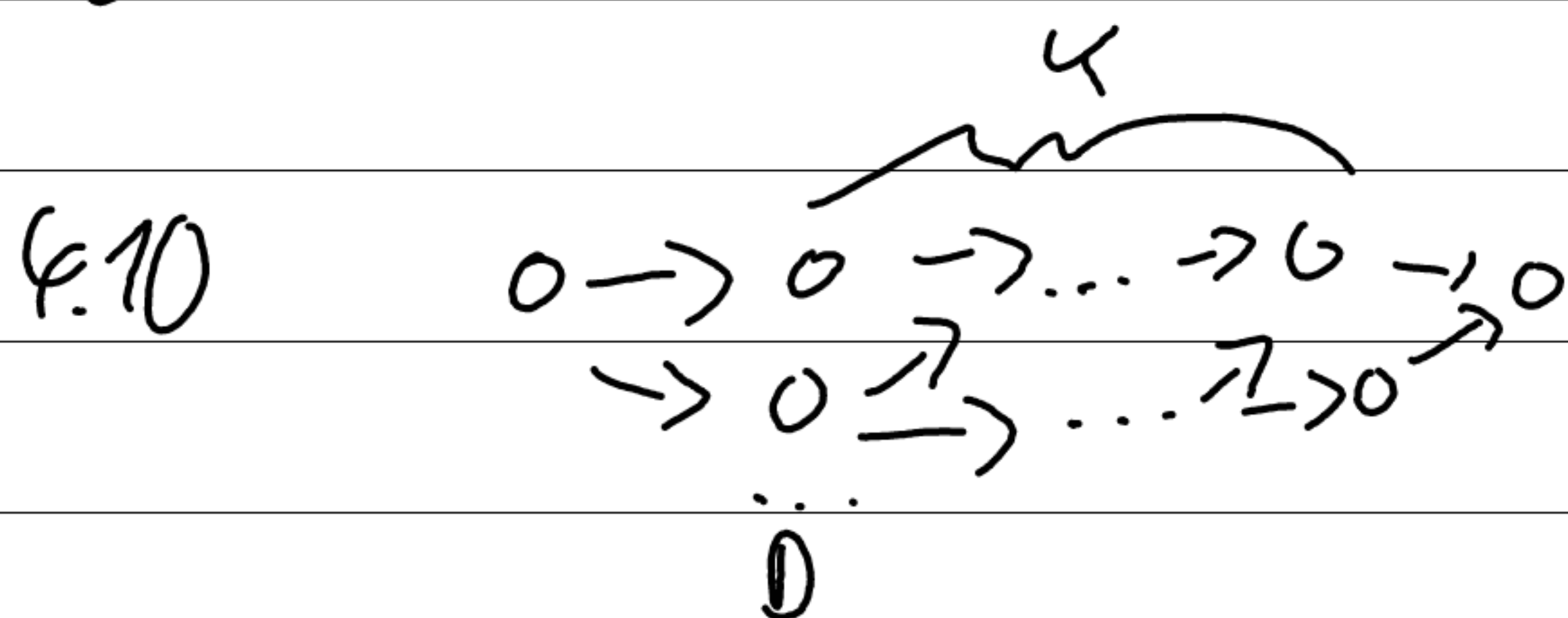
k lines: 4^{k+1} regions (beware each interval uses whole input domain!)

4.9 e.g., $x = \frac{1}{3}$ $x = \frac{2}{3} \leadsto$ slope 3 / -3



nodes only on $[0, 1]$.

$N \geq 3$ we can create regions where the slope = 0 on the right corner!



Input $\rightarrow$ first hidden layer: D weights, D biases: $2D$ parameters

Between hidden layers: $K-1$ transitions, each has D^2 weights and

D biases: $D^2 + D = D(D+1)$, in total $(K-1)D(D+1)$

Last hidden layer $\rightarrow$ output: 1 bias, D weights: $D+1$

$$\Rightarrow \underbrace{2D}_{\text{input}} + \underbrace{(K-1)D(D+1)}_{\text{layers}} + \underbrace{D+1}_{\text{output}}$$

4.17 $D_{in}=1, D_{out}=1$ K layers with D hidden units

$K=10 \wedge D=5$ vs $K=1 \wedge D=95$

param | $2 \cdot 5 + 9 \cdot 5 \cdot 6 + 5 + 1$
= 286

$2 \cdot 95 + 95 + 1$
= 286

linear regions | $6^9 \cdot \left(\binom{5}{0} + \binom{5}{1} \right)$
= $6^9 \cdot 6 = 60466176$

$\binom{95}{0} + \binom{95}{1}$
= $1 + 95 = 96$

Poser? Probably not

Probably (parallel)